where the last equation follows from independence of the increments ($\mathbb{E}Z_k Z_l = 0$ if $k \neq l$).

Also observe that $\Delta \mathrm{W}_k$ is a normally distributed random variable with the following properties

$$\mathbb{E}\Delta \mathrm{W}_k = 0, \qquad \mathbb{E}\Delta \mathrm{W}_k^2 = \delta t = \tau/\nu, \qquad \mathbb{E}\Delta \mathrm{W}_k^4 = 3\delta t^2 = 3(\tau/\nu)^2.$$

Thus we have

$$\begin{aligned} I_\nu &= \mathbb{E}\sum_{k=0}^{\nu-1}(\Delta \mathrm{W}_{\nu,k}^2 - \frac{\tau}{\nu})^2 \\ &= \sum_{k=0}^{\nu-1}\mathbb{E}\left(\Delta \mathrm{W}_{\nu,k}^4 - 2\frac{\tau}{\nu}\Delta \mathrm{W}_{\nu,k}^2 + \frac{\tau^2}{\nu^2}\right) \\ &= \sum_{k=0}^{\nu-1}\left(3\left(\frac{\tau}{\nu}\right)^2 - 2\left(\frac{\tau}{\nu}\right)^2 + \frac{\tau^2}{\nu^2}\right) = 2\frac{\tau^2}{\nu^2}\nu = 2\frac{\tau^2}{\nu}. \end{aligned}$$

Thus $I_\nu \to 0$ as $\nu \to 0$, which is the desired result. □

The calculation implies that

$$\int_0^\tau \mathrm{W}(t)\mathrm{dW}(t) = \frac{1}{2}\left(\mathrm{W}(\tau)^2 - \tau\right).$$

In contrast, the Stratonovich (midpoint) integral (denoted by $\int_0^\tau \circ \mathrm{dW}(t)$) can be calculated as

$$\int_0^\tau \mathrm{W}(t) \circ \mathrm{dW}(t) = \mathrm{l.i.m.}_{\nu\to\infty}\sum_{k=0}^{\nu-1}\mathrm{W}(k\delta t + \delta t/2)[\mathrm{W}((k+1)\delta t) - \mathrm{W}(k\delta t)].$$

Writing

$$\mathrm{W}(k\delta t + \delta t/2) = \mathrm{W}\left(\frac{(k+1)\delta t + k\delta t}{2}\right) = \frac{1}{2}\left(\mathrm{W}((k+1)\delta t) + \mathrm{W}(k\delta t)\right) + \mathcal{O}(\sqrt{\delta t}),$$

allows us to evaluate the integral as

$$\int_0^\tau \mathrm{W}(t) \circ \mathrm{dW}(t) = \frac{1}{2}\mathrm{l.i.m.}_{\nu\to\infty}\sum_{k=0}^{\nu-1}\left[\mathrm{W}((k+1)\delta t)^2 - \mathrm{W}(k\delta t)^2\right] = \frac{1}{2}\mathrm{W}(\tau)^2.$$